

Semantic-Aware Hierarchical 3D Gaussian Representation for Autonomous Driving Simulation

Abstract—3D Gaussian Splatting (3DGS) has recently emerged as an efficient paradigm for high-fidelity urban scene reconstruction in autonomous driving simulation. However, existing methods rely on globally uniform representations, leading to severe redundancy and inefficient resource allocation in large-scale scenes. Moreover, limited viewpoints and occlusions result in insufficient supervision, causing blurry textures and degraded multi-view consistency. To address these challenges, we propose a semantic-aware hierarchical 3D Gaussian representation that jointly optimizes anchor distribution and level-of-detail (LOD) across different semantic regions. Specifically, we introduce a semantic-guided anchor allocation strategy to adaptively control Gaussian density, along with a category-aware LOD mechanism for efficient multi-scale rendering. To further alleviate supervision scarcity, we incorporate diffusion-based pseudo-view augmentation to enrich training signals under occlusions. In addition, a multi-view consistency loss is designed to enforce geometric and photometric alignment across views. Building upon these designs, we establish a complete pipeline from 3D reconstruction to simulation data synthesis. Extensive experiments demonstrate that our method achieves superior rendering quality and geometric accuracy while reducing computational overhead, making it well-suited for autonomous driving simulation.

I. INTRODUCTION

High-fidelity scene reconstruction and simulation play a critical role in the development and validation of autonomous driving systems. Recently, 3D Gaussian Splatting (3DGS) [1] has gained significant attention due to its ability to achieve real-time rendering with high visual fidelity by combining multi-view image information with geometric priors.

Despite these advantages, existing 3DGS-based methods face two fundamental challenges in large-scale urban scenarios. First, most approaches adopt a globally uniform representation, which leads to inefficient allocation of Gaussian primitives—over-representing simple regions while under-fitting complex structures. Second, real-world driving data often suffer from limited viewpoints and severe occlusions, resulting in insufficient supervision that degrades reconstruction quality, especially in terms of texture sharpness and multi-view consistency.

To overcome these limitations, we propose a semantic-aware hierarchical Gaussian representation that enables adaptive and efficient scene modeling. Our key idea is to integrate semantic priors into both spatial distribution and level-of-detail control of Gaussian primitives, forming a unified framework for representation optimization. Specifically, we introduce a semantic-guided anchor allocation strategy to distribute Gaussian primitives according to scene complexity, and a category-aware level-of-detail mechanism to dynamically adjust rendering granularity. Furthermore, to compen-

sate for missing observations, we leverage diffusion-based pseudo-view generation to provide additional supervision signals. A multi-view consistency loss is also designed to enforce alignment across geometry and appearance.

Based on the above designs, we further build a complete pipeline that supports high-quality reconstruction, photorealistic rendering, and flexible simulation data generation for autonomous driving applications.

The main contributions of this work are summarized as follows:

- We propose a semantic-aware hierarchical Gaussian representation that jointly optimizes anchor distribution and level-of-detail for efficient large-scale scene reconstruction.
- We propose a generative prior-enhanced multi-view consistent weakly supervised reconstruction method and design a multi-view consistency loss to improve geometric and photometric alignment across views.
- We establish a complete reconstruction-to-simulation pipeline and demonstrate its effectiveness on multiple autonomous driving datasets.

II. RELATED WORK

A. 3D Gaussian Splatting

3DGS has emerged as a breakthrough in 3D scene reconstruction and visualization, enabling high-fidelity, real-time radiance field rendering. It outperforms traditional methods like NeRF [2] in real-time performance while maintaining comparable or higher fidelity, thus gaining widespread attention. Standard 3DGS represents scenes using parameterized 3D Gaussians, enabling photorealistic real-time rendering via projection and blending. However, its global uniform representation causes inefficiency and redundancy in large-scale urban driving scenes, wasting computing resources by allocating equal Gaussian density to simple and complex regions. The lack of adaptive adjustment also hinders balancing reconstruction accuracy and resource efficiency, limiting its use in resource-constrained autonomous driving scenarios.

B. Scene Representation for Autonomous Driving

High-quality and efficient 3D scene representation is essential for autonomous driving simulation, where both geometric fidelity and real-time performance are critical for downstream perception and planning tasks. Traditional point-based methods rely on LiDAR priors to recover scene geometry with high precision, but often produce sparse and incomplete structures in occluded or texture-less regions,

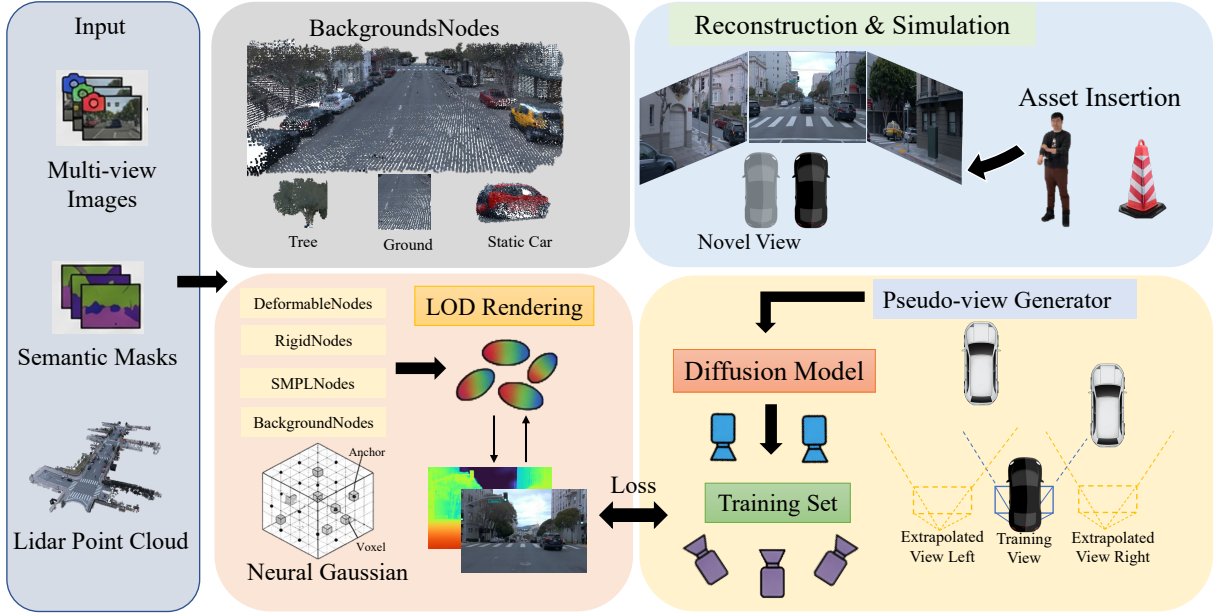


Fig. 1. Overview of our method. We take multi-view RGB images, semantic masks, and LiDAR point clouds as input. We initialize category-aware anchor points from background point clouds to construct neural 3D Gaussians, and apply semantic-aware LOD strategies to balance efficiency and accuracy. A diffusion model generates pseudo-views to enrich supervision and reduce overfitting. Our framework achieves high-precision 3DGS reconstruction, photorealistic free-viewpoint rendering, and flexible asset insertion for realistic autonomous driving simulation.

failing to provide dense and photorealistic appearance rendering. Mesh-based representations achieve compact storage and efficient rendering, yet struggle with complex topological structures and dynamic objects in urban traffic scenarios, leading to noticeable geometry distortion and detail loss. 3DGS-based approaches have enabled high-fidelity novel view synthesis with impressive rendering speed. However, most existing frameworks adopt a globally unified representation paradigm, which leads to severe redundancy in large-scale scenes and imbalanced allocation of modeling resources. Simple background regions are over-represented while intricate dynamic objects and semantic boundaries remain under-optimized, resulting in degraded reconstruction quality. Although recent works introduce semantic cues to guide scene modeling, they still lack a hierarchical and category-adaptive mechanism to dynamically adjust level-of-detail. As a result, nearly all existing methods face an inherent trade-off: either sacrificing realism for efficiency or consuming excessive computation for fidelity. Few approaches simultaneously achieve high reconstruction quality, semantic awareness, and resource efficiency, making them difficult to deploy in real-world autonomous driving systems that demand both reliable simulation and low-latency inference.

III. METHOD

A. Overview

Our goal is to reconstruct a high-fidelity and efficient 3D Gaussian representation for autonomous driving scenes from multi-view RGB images and LiDAR point clouds. As illustrated in Fig. 1, we address two key challenges: (1) inefficient global representations in large-scale scenes, and

(2) insufficient supervision caused by occlusions and limited viewpoints.

To this end, we propose a **semantic-aware hierarchical Gaussian representation** that unifies spatial distribution and level-of-detail (LOD) optimization. The overall framework consists of three components: 1) semantic-aware hierarchical representation for efficient modeling, 2) diffusion-based pseudo-view augmentation for enhanced supervision, and 3) multi-view consistency optimization for robust reconstruction.

B. Semantic-Aware Hierarchical Gaussian Representation

1) *Semantic-Guided Anchor Allocation*: Instead of using a globally uniform Gaussian distribution, we allocate Gaussian primitives based on semantic structure to better match scene complexity. Specifically, we partition the static background into semantic groups:

$$G = \{G_{\text{ground}}, G_{\text{tree}}, G_{\text{car}}, G_{\text{other}}\}. \quad (1)$$

Dynamic objects are excluded using bounding boxes to obtain a clean background point cloud. For each semantic group g , we assign a category-specific density coefficient α_g , which controls the number of Gaussian anchors:

$$N_g = \max(\lfloor N_{\text{base}} \cdot \alpha_g \rfloor, 1). \quad (2)$$

Intuitively, complex and high-frequency regions (e.g., trees) are allocated more anchors, while smooth regions (e.g., ground) are sparsely represented, improving efficiency without sacrificing detail.

We then apply Farthest Point Sampling (FPS) to generate anchors:

$$A_g = \text{FPS}(P_g, N_g), \quad (3)$$

which ensures spatial coverage and avoids clustering compared to uniform or voxel-based initialization. The final anchor set is:

$$A = \bigcup_g A_g. \quad (4)$$

2) *Category-Aware Level-of-Detail (LOD)*: To further improve efficiency, we introduce a semantic-aware LOD mechanism that adaptively controls rendering granularity.

For each anchor a_i , we compute a normalized distance ratio:

$$\bar{d}_i = \min\left(\frac{d_i}{d_{\max}}, 1\right), \quad (5)$$

and define a semantic-aware effective distance:

$$\hat{d}_i = \beta_{l_i} \cdot \bar{d}_i, \quad (6)$$

where β_{l_i} is a category-specific multiplier. This design assigns larger effective distances to less critical regions (e.g., ground), encouraging early pruning of fine details while preserving important structures.

The maximum LOD level is then computed as:

$$k_i^* = \left\lfloor (1 - \hat{d}_i) \cdot k_{\max} \right\rfloor. \quad (7)$$

During rendering, only anchors satisfying $k_i \leq k_i^*$ are activated, enabling coarse-to-fine adaptive rendering.

3) *Dynamic Anchor Optimization*: To maintain a compact yet expressive representation, we perform dynamic anchor refinement during training. When splitting anchors, child anchors inherit semantic labels and progressively increase detail levels:

$$k_{\text{child}} = \min(k_p + 1, k_{\max}). \quad (8)$$

Conversely, anchors with low contribution are pruned. Importantly, semantic labels and LOD levels are updated synchronously to ensure consistency across attributes. This mechanism enables scalable optimization while preserving structural coherence.

C. Diffusion-Based Pseudo-View Augmentation

To alleviate supervision scarcity caused by occlusions and sparse viewpoints, we introduce a generative prior-guided pseudo-view augmentation strategy.

Specifically, we first generate novel viewpoints by interpolating along camera trajectories. These views are then refined using a diffusion model to produce geometrically and semantically consistent images. The generated pseudo-views are progressively incorporated into the training set.

This strategy effectively enriches supervision signals in under-observed regions and reduces overfitting to limited viewpoints, leading to improved reconstruction completeness and robustness.

D. Multi-View Consistency Optimization

To enforce consistency across views, we design a multi-view consistency loss that jointly constrains geometry and appearance.

The overall training objective is:

$$\mathcal{L} = (1 - \lambda_r)\mathcal{L}_1 + \lambda_r\mathcal{L}_{\text{SSIM}} + \lambda_d\mathcal{L}_{\text{depth}} + \lambda_o\mathcal{L}_{\text{opacity}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{mv}} \quad (9)$$

The multi-view consistency loss is defined as:

$$\mathcal{L}_{\text{mv}} = \lambda_{\text{geo}} \frac{1}{|M|} \sum_{p \in M} w_p \|p - \hat{p}\|_2^2 + \lambda_{\text{ncc}} \frac{1}{|M|} \sum_{p \in M} w_p (1 - \text{LNCC}(I_1(p), I_2(p))) \quad (10)$$

where the first term enforces geometric consistency via reprojection error, and the second term enforces photometric consistency using normalized cross-correlation. The adaptive weight w_p suppresses outliers based on reprojection uncertainty.

E. Simulation-Oriented Scene Construction

Building upon the reconstructed Gaussian representation, we further support flexible asset insertion for autonomous driving simulation. Pre-optimized Gaussian assets (e.g., pedestrians, vehicles) can be seamlessly integrated into the scene while maintaining geometric and photometric consistency as shown in Fig. 2.

This enables the generation of diverse and photorealistic simulation data, facilitating downstream perception and planning tasks.



Fig. 2. Asset Insertion Examples. We provide four categories of assets (car, obstacle, human [3], animal) for autonomous driving scene augmentation, which can be inserted into 3D reconstructed scenes to simulate real-world driving environments.

IV. EXPERIMENTS

A. Datasets and Evaluation Metrics

We evaluate our method on three widely used autonomous driving datasets: Waymo Open Dataset [4], KITTI [5], and nuScenes [6], covering diverse urban environments and sensor configurations. Waymo is used as the primary benchmark for quantitative and qualitative evaluation, while KITTI and nuScenes are adopted to validate cross-dataset generalization.

Following prior works, we evaluate rendering quality using Peak Signal-to-Noise Ratio (PSNR) [7], Structural Similarity Index (SSIM) [8], and Learned Perceptual Image Patch



Fig. 3. Qualitative results on Waymo dataset

TABLE I

QUANTITATIVE COMPARISON ON WAYMO, KITTI AND NUSCENES DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Model	Waymo			Kitti			Nuscenes		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
PVG	29.42	0.86	0.22	26.22	0.89	0.06	28.29	0.82	0.25
OmniRe	31.16	0.90	0.18	28.18	0.91	0.08	28.38	0.85	0.22
Ours	32.52	0.92	0.15	29.77	0.92	0.05	28.67	0.84	0.24

Similarity (LPIPS) [9]. Higher PSNR and SSIM indicate better reconstruction fidelity, while lower LPIPS reflects improved perceptual similarity. In addition, we report GPU memory consumption and training time to assess computational efficiency.

B. Implementation Details

Our model is trained for 30k iterations using multi-view RGB images and LiDAR point clouds. The training process is divided into two stages: an initial reconstruction stage and a pseudo-view augmented stage. Diffusion-based pseudo-views are progressively incorporated after 20k iterations to enhance supervision. We set the maximum LOD level $k_{\max} = 4$ and the maximum observation distance $d_{\max} = 100$. Category-specific density coefficients α_g and distance multipliers β_{l_i} are empirically determined based on scene complexity. All experiments are conducted on a single GPU A800.

C. Baselines

We conducted comprehensive comparative experiments between our proposed method and several state-of-the-art 3D Gaussian-based methods on the Waymo dataset. To verify cross-scenario generalization, we also evaluate on KITTI and nuScenes. We compare with StreetGS [10], PVG [11], BezierGS [12], and the strong baseline OmniRe [13].

D. Comparison with State-of-the-Art

Quantitative Results. As shown in Table I and Table II, our method consistently outperforms all baselines across multiple datasets. On the Waymo dataset, our approach achieves a PSNR of 32.52 and an SSIM of 0.92, surpassing OmniRe (31.16/0.90) and StreetGS (31.96/0.92). Our method

TABLE II

QUANTITATIVE COMPARISON ON WAYMO DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Model	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
StreetGS	31.96	0.92	0.14
BezierGS	25.23	0.84	0.28
PVG	29.42	0.86	0.22
OmniRe	31.16	0.90	0.18
Ours	32.52	0.92	0.15

also achieves the lowest LPIPS, indicating better perceptual realism.

Qualitative Results. As illustrated in Fig. 3, our method generates sharper object boundaries, clearer textures, and more consistent geometry than baseline methods. In occluded and complex regions, our approach effectively reduces blurring and distortion. On KITTI and nuScenes, our method remains competitive or superior, showing strong generalization ability under different data distributions and sensor setups as shown in Fig.4.

E. Efficiency Analysis

We analyze efficiency in terms of GPU memory usage and training time (Table III). Our method reduces memory consumption compared to OmniRe while achieving higher reconstruction quality, owing to the semantic-aware LOD strategy that avoids redundant Gaussians in simple regions. This enables a better trade-off between quality and computation for large-scale scenes.

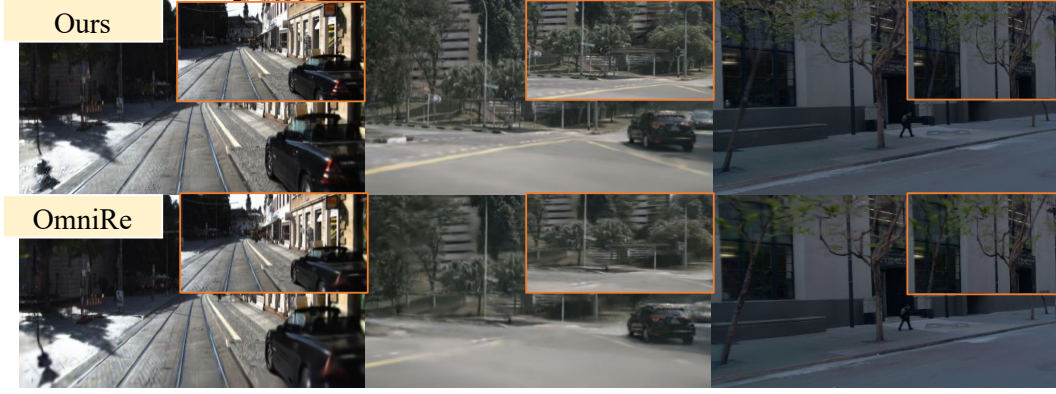


Fig. 4. Qualitative comparison of rendering results between our method and OmniRe on KITTI (left), nuScenes (middle), and Waymo (right) datasets. The zoomed-in regions highlight that our method produces sharper object details, clearer texture, and more accurate geometric structure compared to OmniRe, especially in challenging street scenes with complex lighting and dynamic objects.

TABLE III

QUANTITATIVE COMPARISON OF PERFORMANCE AND EFFICIENCY, AND ABLATION STUDY ON KEY COMPONENTS.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	GPU Memory (GB)	Training Time
OmniRe	29.73	0.88	0.21	18.19	42min
Ours (Full)	31.72	0.92	0.14	14.94	48min
w/o category-aware LOD	30.24	0.89	0.17	18.34	46min
w/o Diffusion Augmentation	30.81	0.90	0.18	14.63	31min
w/o Multi-view Loss	30.12	0.88	0.19	14.82	46min



Fig. 5. Visualization of rendering results during training. We compare our method and OmniRe at different training steps. As training progresses, our method achieves clearer scene details, sharper object edges, and more accurate geometric structure compared to OmniRe, demonstrating faster convergence and better rendering quality.

F. Ablation Study

To validate each core component, we perform ablation experiments on the Waymo dataset (Table III). Removing any module leads to clear performance drops, with increased blurriness and structural inconsistency.

Ablating semantic-aware anchor allocation results in fragmented scene structures. Disabling category-aware LOD increases redundancy and degrades fine details. Removing diffusion-based pseudo-view augmentation leads to unstable textures and incomplete reconstruction in occluded areas. Removing multi-view consistency loss causes geometric misalignment and view inconsistency. Our full model achieves the best overall performance.

Asset Insertion & Novel View

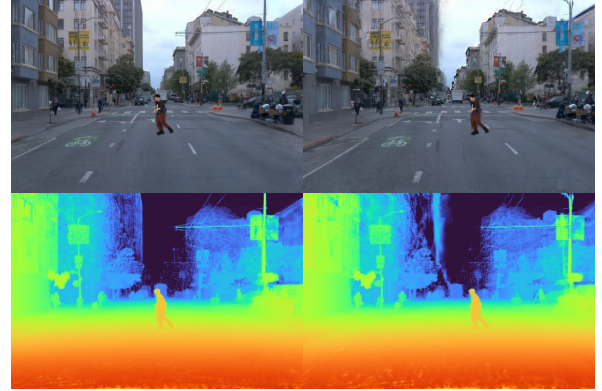


Fig. 6. Qualitative demonstration of asset insertion and novel view rendering. Top row: Rendered RGB images of a street scene with an inserted human asset, viewed from original view and novel view. Bottom row: Corresponding depth maps, which verify the accurate geometric consistency of the inserted asset with the 3D scene structure across novel views. Our method enables photorealistic asset insertion with correct occlusion, lighting, and perspective.

G. Training Dynamics Visualization

We visualize the training process in Fig. 5. Compared with OmniRe, our method converges faster and produces clearer structures at earlier iterations, benefiting from better representation design and enhanced supervision.

H. Simulation-Oriented Asset Insertion

To enrich simulation scenarios, we leverage SAM3D [14] and GART [15] to obtain high-quality Gaussian assets including vehicles, pedestrians, and obstacles. As shown in Fig. 6, our framework supports seamless asset insertion with consistent geometry, appearance, and occlusion relations, enabling flexible and photorealistic autonomous driving simulation data generation.

V. CONCLUSION

Large-scale outdoor autonomous driving scene reconstruction suffers from redundant 3D Gaussian representations

and degraded rendering quality, often manifested as blurry textures and loss of geometric details. These issues are further exacerbated by occlusions and insufficient multi-view consistency. To address these challenges, this paper proposes a semantic-aware anchored 3D Gaussian splatting framework. By employing semantic-driven adaptive anchor assignment and category-aware level-of-detail strategies, the method achieves a fine-grained balance between rendering quality and computational efficiency. Through generative prior enhancement and multi-view consistency constraints, it mitigates the issue of insufficient supervision for occluded objects. Furthermore, this work establishes a complete closed-loop pipeline spanning scene reconstruction, synthetic data generation, and perception validation. Experimental results demonstrate that the proposed approach outperforms mainstream baseline methods across multiple metrics, maintaining high reconstruction quality while preserving efficiency. In addition, the high-quality synthetic data generated by our method effectively improves the performance of autonomous driving perception models, conferring this work significant academic value and practical implications for real-world engineering deployment.

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